



Operations Research

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Exercise Sheet 11

Training Exercises

Exercise T11.1. *Convex Functions - Examples*

Check whether the following functions are convex.

- a) $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^4 - x$
- b) $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = |x| + \cos(x)$
- c) $\max \{f(x), g(x)\}$ for $f, g : \mathbb{R}^n \rightarrow \mathbb{R}$ convex
- d) $f : \mathbb{R}^n \rightarrow \mathbb{R},$

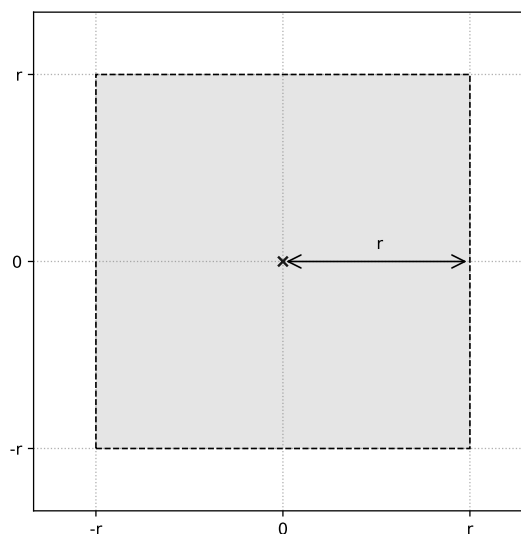
$$f(x) = \begin{cases} \|x\| & \text{if } \|x\| \leq 1 \\ \|x\|^2 & \text{else} \end{cases}$$

Exercise T11.2. *Convex functions*

Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R},$

$$f(x, y) = \exp(2x) + xy + y^2.$$

For $r > 0$, denote with $Q_r = \{(x, y) \in \mathbb{R}^2 : \max(|x|, |y|) < r\}$ the square around $(0, 0)$ with side length $2r$, (see Figure below).



Give an $r > 0$ such that $f : Q_r \rightarrow \mathbb{R}$ is convex. Justify your answer.

Exercise T11.3. *The Hedge Algorithm and the Learning Rate*

Consider an online prediction problem with $K = 2$ experts. Initialize the weights uniformly as $w_1^{(1)} = 1$ and $w_2^{(1)} = 1$. The observed loss vector in round 1 is $L^{(1)} = (0.2, 1.0)^T$.

- (a) Let the learning rate be $\eta = 1$. State the multiplicative weight update rule and compute the updated probabilities $p_1^{(2)}$ and $p_2^{(2)}$ for the second round.
- (b) The expected loss of the algorithm in any round t is defined as $\mathbb{E}[L^{(t)}] = \sum_{i=1}^K p_i^{(t)} L_i^{(t)}$. Compute the expected loss of the algorithm in round 1. How does it compare to the loss of the best expert in round 1?
- (c) Consider the general update rule parameterized by $\eta > 0$. Analyze the mathematical limits of the updated probability distribution $p^{(2)}$ as $\eta \rightarrow 0$ and as $\eta \rightarrow \infty$. Discuss what these limits imply about the algorithm's behavior regarding exploration versus exploitation.